30/04

Passing object to a function and returning object from a function

Complex ----> a+bi
real,img
Add
sub
Mul
SetData
Print

syntax to define member function outside the class

return datatype classname:: func()
{

}

ex : void Complex :: SetData()
{

}

```
int x=10;
int a[]={10,20,30,40};
struct St V={123,"abcd"};

Complex e1;
Rectangle robj;
```

Constructor

A constructor is a special member function which is used to initialise the object at the time of its declaration

Points to be noted:

1. Constructor name should be same as class name
2. Constructor must be public function
3. Constructor should not have any return type not even void should be mentioned

Types of Constructors

We have 3 types of constructors

1. default constructor
2. parameterised constructor
3. copy constructor

Default constructor : A constructor without input parameters ot input arguments is called default constructor

Note : An uninitialised object always invokes default constructor.

ex : Complex e1; /// e1.Complex();

When none of the constructor are supplied by the programmer, compiler will supply dummy default constructor for name sake.

A constructor with input paramters is called parameterised constructor.

Note : An object with inputs or parameters will invoke parameterised constructor

ex : Complex e2(56,78); // e2.Complex(56,78);

A constructor whose input is same class object reference is called copy constructor.

Note : Copy constructor is invoked when we assign existing object to newly created object in its declaration

ex : Complex e1;
Complex e2=e1; /// e2.Complex(e1);

Complex(Complex &e)
{
}